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Integrating Passive Acoustics And Thermal Modeling To Optimize Remedial Strategies For Controlling Water Influx And Enhancing Oil Recovery

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Abstract

Water cut, defined as the percentage of water in the total produced fluid, is a common and growing challenge in aging oil reservoirs. As reservoirs mature, water production typically increases due to several factors, including declining reservoir pressure, coning from underlying aquifers, water channeling through high-permeability streaks, poor zonal isolation, and breakthrough from water injectors. These factors pose significant challenges to efficient and economical oil extraction. Therefore, effective water cut management is critical for sustaining reservoir performance and long-term field profitability. This requires continuous monitoring and strategic interventions to mitigate serious impacts on hydrocarbon production.

Conventional diagnostic tools, such as spinners, are limited to detecting flow within the wellbore coming from perforated intervals [1]. However, in wells with increasing water cut, the actual source of water influx can originate outside the perforated zones, above or below, due to the presence of an active aquifer and pressure differentials between formations. Additionally, as cement integrity deteriorates over time, the development of channels and micro-annulus in the cement may allow fluid to migrate through unintended pathways, bypassing designed isolation barriers and complicating standard diagnostic techniques. This highlights the need for integrated, high-resolution logging tools capable of identifying crossflow issues beyond the perforated sections, thereby enabling more effective and lasting remediation strategies.

This paper presents an advanced production logging tool (PLT) approach that integrates conventional spinner measurements with passive acoustics (Chorus) and high-precision temperature (HPT) logging to accurately identify the source of water breakthrough in one of the wells in a mature oil field in Libya. Temperature modeling (Cascade) was used to quantify the volume of behind-casing flow and guide the development of a targeted remedial action plan. This plan involved a water shut-off cement squeeze operation aimed at sealing channels within the degraded cement behind the casing, in order to isolate the source of water influx and enhance oil recovery. Following the intervention, a 40% reduction in water cut was observed, confirming the success of the operation.

Keywords: Water Breakthrough Detection, Cement Squeeze Evaluation, Chorus Passive Acoustics Technology, Cascade Temperature Modelling Technology, Production Logging

Introduction

The surveyed well was drilled in February 2006 as a producer well. In May 2006, a Y-tool and electric submersible pump (ESP) were installed, and the well was subsequently brought online in January 2007. Initial production showed a water cut of 50%, which was unexpected given the well's location high on the structural peak of the field.

Over the next months, the water cut increased to 68%, prompting investigations into the source of water ingress. Two PLTs were conducted to investigate water entry into the well. The first, in October 2007, found that the lowest perforated interval was the main source of water. The second, in September 2009, showed water entering through the two lowest perforations.

Salinity analyses of the produced water consistently confirmed the presence of formation water, with no indication of breakthrough from injector wells. In August 2021, a new ESP was installed, and the well was returned to production. However, a routine well test soon revealed an unexpected decline in oil production accompanied by a rise in water cut, triggering further analysis. A subsequent workover was performed in September 2022. As part of this operation, a casing integrity test was conducted, which confirmed that there were no leaks in the casing that could allow water from the shallow aquifer to enter the wellbore.

Despite these interventions, high water production persisted. Post-workover wellhead liquid sampling revealed total dissolved solids (TDS) levels inconsistent with the target reservoir, suggesting external water source. A re-evaluation of the 2009 PLT data supports the hypothesis that water entry at the lower perforations is likely caused by behind-casing flow from deeper zones, as indicated by the temperature response behaviour.

During the production phase, reservoir pressure declines, disrupting the initial pressure balance. Reservoirs with ongoing production experience pressure depletion, while non-producing zones maintain their original pressure. This differential pressure can initiate inter-reservoir communication, as fluids naturally migrate along the path of least resistance. Over time, degradation of cement integrity may further facilitate flow between formations by the channeling behind casing. In such scenarios, underlying aquifers can become a potential source of water breakthrough and contribute to high water cut.

Methods

To confirm the source of water production, a combination of passive acoustics and high-precision temperature logging was carried out (Fig. 1). Temperature modeling was then used to estimate the volume of behind-casing flow to support the design of a targeted remedial action plan. These efforts contribute to a more comprehensive understanding of fluid movement within both the wellbore and the reservoir.

Temperature data plays a vital role in oil and gas well diagnostics, offering valuable insights into zonal contribution in production and injection wells [2]. These interpretations can be further validated through numerical simulation of the temperature profiles to support accurate flow quantification [3]. A tailored logging program was developed to quantify behind-casing flow using Cascade temperature modelling technology. Data acquisition was performed under shut-in, flowing, and transient conditions (Fig. 1). A key advantage of Cascade temperature modelling technology in this context is its ability to identify flow zones behind unperforated casing, potentially linked to active reservoir units or crossflow between formations caused by behind-casing channeling.

Fluid movement produces acoustic signals across a range of frequencies and intensities. The noise intensity increases with higher flow rates and greater differential pressures, while the frequency content is inversely related to the size or aperture of the flow path [4]. For instance, flow through smaller pores or narrow channels generates higher-frequency noise, whereas flow through larger pores produces lower-frequency signals. This relationship forms the basis of passive acoustic logging (Chorus), enabling the tool to differentiate between the different sources and pathways of fluid movement [4].

The purpose of logging under shut-in conditions was to record a baseline temperature profile and detect any active crossflow under static conditions using passive acoustic measurements. During the flowing mode, the objective was to identify zones actively contributing to flow. The presence of the electric submersible pump (ESP) in the surveyed well introduced several operational challenges for the logging, primarily due to the requirement to perform measurements through a Y-tool. The restricted internal diameter and complex geometry of the Y-tool can limit the types and sizes of logging tools that can be deployed, while also complicating tool conveyance. Furthermore, logging in the presence of an active ESP introduces additional thermal heating and acoustic noise that can interfere with temperature and passive acoustics data quality.

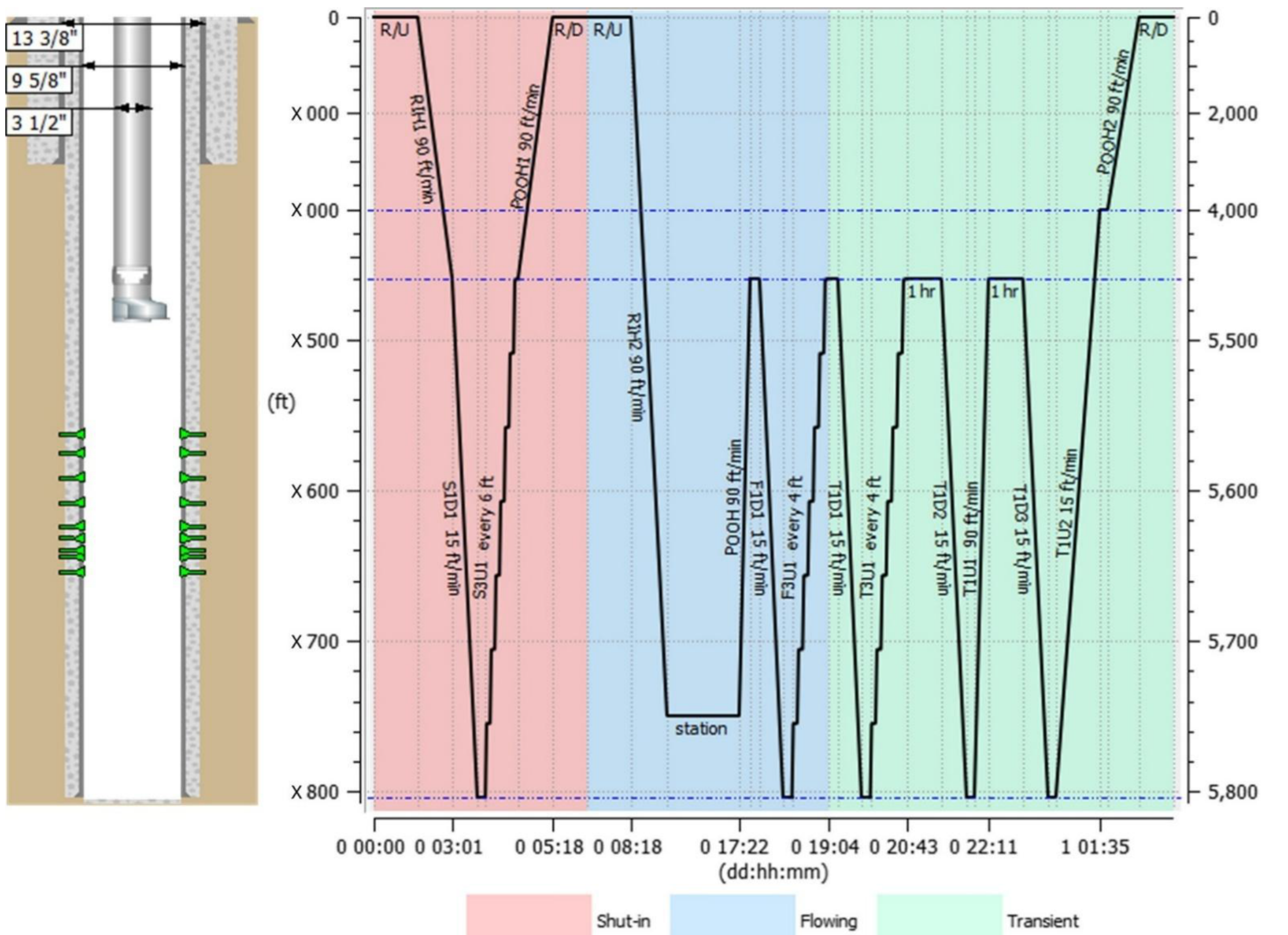

Figure 1—Logging Procedure

To minimize the thermal influence of the ESP on the logging tools during the six-hour flow stabilization period (Fig. 1), the stabilization station was intentionally placed at the bottom of the well, below the perforated zones. This positioning also reduced the tools' exposure to vibrations and flow-related disturbances. As a result, temperature and pressure were able to stabilize before the flowing logging passes, and the direct impact of pump-generated heat on temperature measurements was significantly reduced.

Multiple transient passes were recorded to analyze the temperature relaxation back to the initial geothermal profile. Additionally, acoustic data acquisition during the transient phase was planned to assess residual reservoir activity, which might be masked during the flowing stage due to the active operation of the ESP pump. Careful planning of logging intervals and flow conditions was essential to ensure data reliability in this complex environment.

Results

Based on passive acoustic logging (Chorus) data recorded under flowing conditions, four primary contributing intervals were identified within the logged section: three across the perforations (Fig. 2 - indicated in green) and one located below the logging interval (Fig. 2 - shown in red). A shallow acoustic streak observed across the ESP (Fig. 2 - highlighted in yellow) was attributed to the operating pump. The absence of acoustic signals during shut-in and transient conditions suggests there is no active crossflow under static conditions.

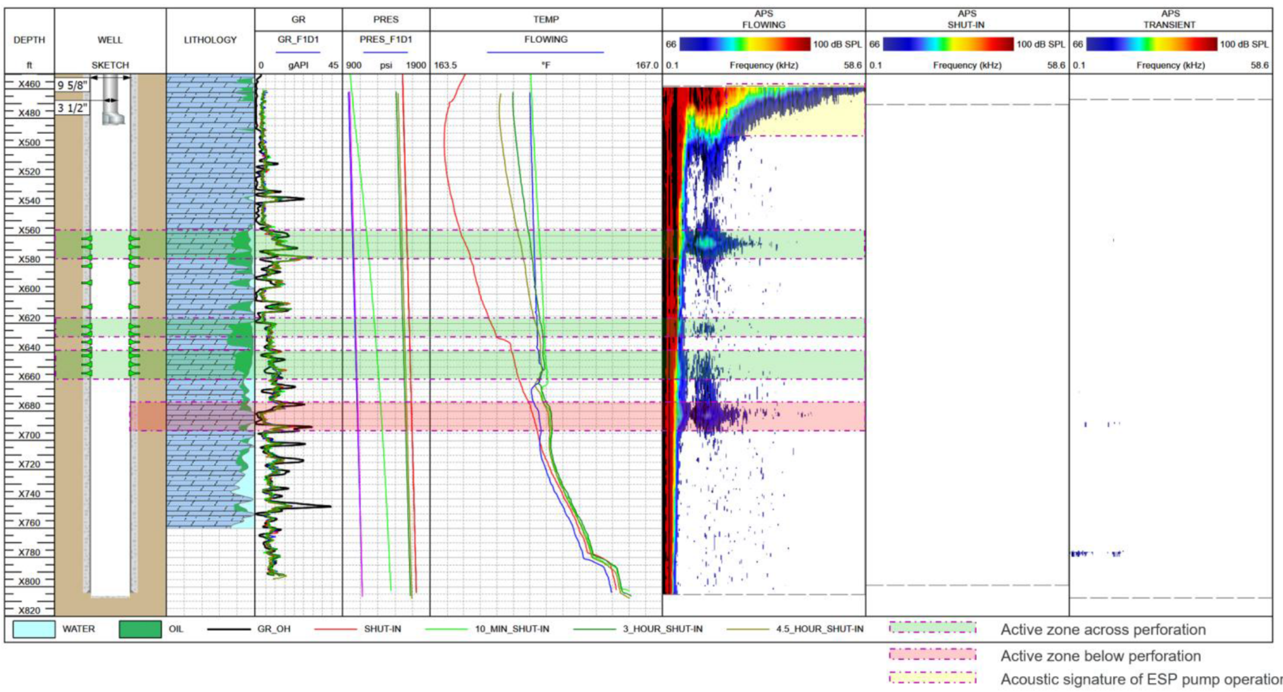

Figure 2—Logging Results

Additionally, the temperature difference between the shut-in and flowing conditions below the lower perforation, together with the presence of low-frequency noise, suggests behind-casing channeling from unperforated reservoirs located deeper. The differential pressure between the static and flowing surveys across the top of the perforation was approximately 600 psi (Fig. 2).

A temperature simulation was conducted to gain quantitative insights and develop a comprehensive understanding of the reservoir's performance. The modelling process incorporated high precision temperature data recorded during both flowing and shut-in conditions, along with critical input parameters such as production history, reservoir parameters, PVT data, etc. Active reservoir units, identified through passive acoustic logging (Chorus), were used directly to define reservoir thickness and contributing zones within the hydrodynamic model. This integrated approach enabled precise calibration of the thermal and hydrodynamic models, enhancing the accuracy of reservoirs behavior assessment and contribution to the production.

In contrast to traditional techniques such as production logging tools (PLT), the use of advanced acoustic and thermal diagnostics offers a significantly broader and more sensitive scanning range. This enhanced diagnostic capability allows for the detection and analysis of complex flow dynamics, including crossflows, channeling behind casing, and flow from unperforated or low-permeability zones, that are often missed by conventional tools due to resolution limitations or mechanical restrictions. As a result, this approach provides a more complete and accurate picture of the reservoir, making it an invaluable tool for optimizing production strategies, identifying inefficiencies, and managing reservoirs with complex geological structures.

Temperature modeling results identified the primary water production zone (~45%) below the perforation intervals (Fig. 3). Open-hole (OH) and saturation logs further confirmed that the formations below the perforated section are predominantly water-bearing. The strong correlation between the modeling outcomes and log data strengthens this interpretation and confirms the presence of water-dominant zones in the lower side of the reservoir. Based on the modelling results the majority of oil production (~40%) was attributed to the bottom perforation interval, aligning with PLT data, which indicates approximately 85% across the bottom perforation, which is actually a commingled liquid production, including contribution from the reservoirs located below (Fig. 3).

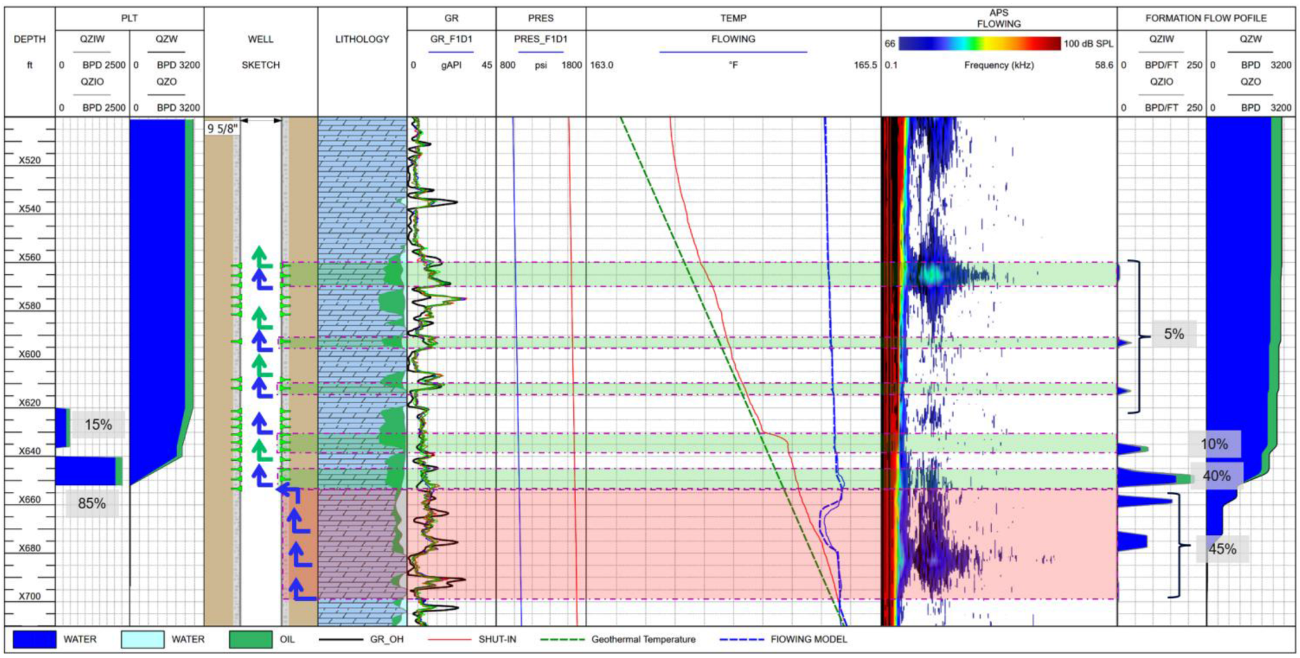

Figure 3—Cascade Modelling Results vs PLT

The temperature modeling also revealed valuable inflow information from the middle and upper perforations, which had not been captured by the spinner tool, likely due to threshold limitations, tool sensitivity issues, or partial turbine blockage (Fig. 3). This demonstrates the advantage of integrating passive acoustic and temperature logging data, which can detect minor flow contributions that are otherwise overlooked. By overcoming the limitations of mechanical tools, this methodology enhances the resolution and reliability of zonal contribution analysis.

One of the main objectives of the logging was to find the source of water breakthrough. Identifying this water-producing interval is critical for optimizing reservoir performance. It enables targeted water management strategies, such as selective zonal isolation, water shutoff treatments, or recompletion decisions, to minimize water cut and maximize hydrocarbon recovery. Understanding the distribution, pressure behavior, and mobility of water-bearing formations helps refine production forecasts and reservoir models, ultimately improving both short-term productivity and long-term recovery efficiency. Furthermore, these insights provide valuable guidance for future field development planning and completion design, especially in mature or water-challenged reservoirs.

Combining acoustic logging with high-precision temperature profiling offers a more complete view of reservoir performance. It allows operators to better understand the contribution of each zone, the interaction between different layers, and the evolution of production over time. These detailed insights are essential for developing targeted, data-driven remediation plans and optimizing well performance.

Data Integration and Remediation

Comprehensive data acquired from passive acoustics and Cascade modelling provided critical insights into the specific contribution zones within the reservoir and precisely identified the sources of excess water production. These detailed diagnostics were instrumental in shaping a focused remediation strategy, enabling a targeted and efficient resolution to the issue.

Based on this analysis, a cement squeeze operation was recommended to effectively halt water flow behind the casing, addressing the root cause of the water influx. This intervention aimed to improve the well's production profile and support more efficient reservoir management.

By eliminating behind-casing water movement, the successful cement squeeze operation can not only reduce water production but also enhance oil recovery, thereby improving overall operational efficiency. This targeted intervention is particularly valuable for mitigating the long-term effects of unwanted water influx on well performance and promoting more sustainable hydrocarbon extraction.

Passive acoustic logging is a powerful diagnostic tool capable of detecting active fluid movement within a 3 to 5-meter radius, regardless of the wellbore completion type [5]. In this case, it was used to identify crossflow behind the casing by capturing acoustic signals generated by fluid migration through the channel. Such crossflow may result from fractured cement or a compromised reservoir barrier [6]. It is typically characterized by clearly defined upper and lower boundaries. In the Chorus acoustic spectrum, it appears as a narrow vertical band linking two active flow zones, a distinctive channeling signature (Fig. 4).

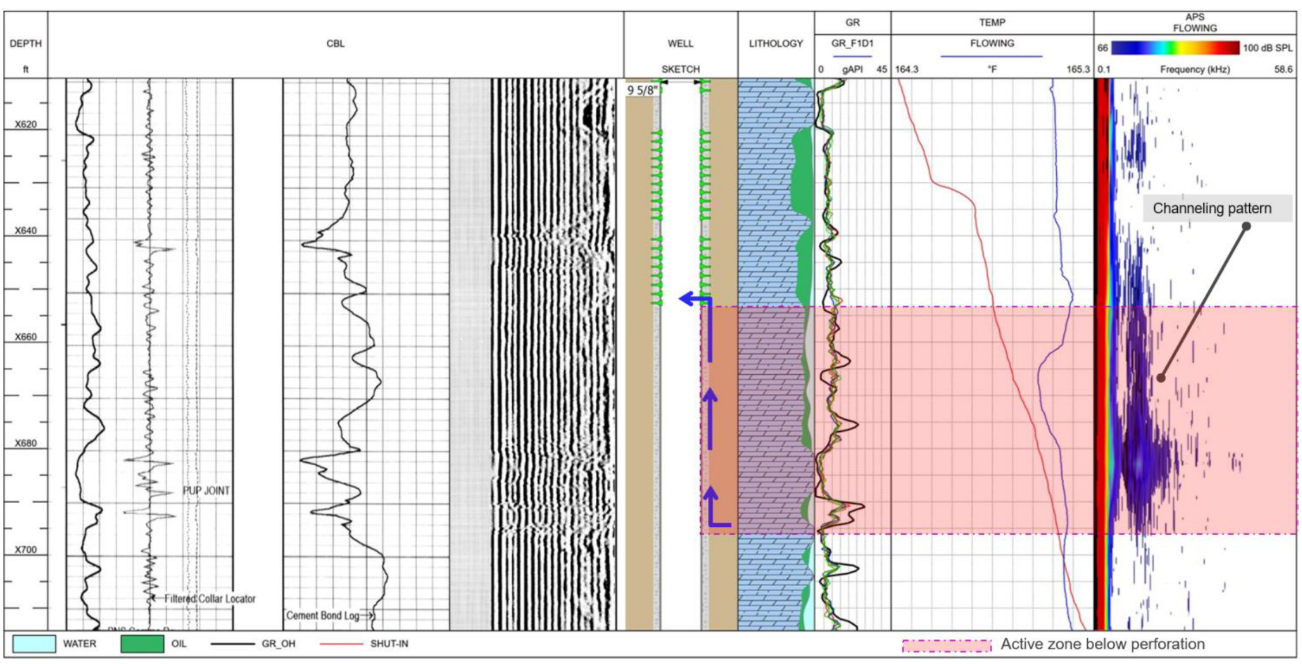

Figure 4—Passive Acoustics / HPT Logging Results vs CBL

When combined with cement bond log (CBL) data, acoustic data also provided critical insights into cement integrity, revealing anomalies such as choke points. These insights were instrumental in designing a more effective and precise cement remediation strategy (Fig. 4).

Following the diagnostic results that successfully identified the source of the water influx, it was decided to implement a cement squeeze job. A cement squeeze is a well remediation technique designed to seal off unwanted fluid flow behind the casing or within the formation. This process involves injecting a specially formulated cement slurry under pressure into targeted zones where leaks or water production occur. The slurry is forced into fractures, channels, or voids behind the casing or within the reservoir rock, creating an impermeable barrier.

By isolating these problematic pathways, the cement squeeze restores zonal isolation, preventing further unwanted water ingress and significantly improving both well integrity and production performance. The operation typically requires precise diagnosis to locate the flow pass and determine its extent, followed by controlled injection to ensure effective sealing without damaging the formation or wellbore.

Successful execution of a cement squeeze relies heavily on the accurate identification of affected zones through integrated diagnostic techniques such as passive acoustic logging, cement bond logs, and temperature analysis. These diagnostics provide critical insights into the location and nature of fluid channels, enabling targeted treatment that minimizes the risk of incomplete isolation or damage to productive intervals.

As a cost-effective intervention, cement squeezes are widely employed in both onshore and offshore operations, showcasing their versatility and reliability across diverse reservoir conditions.

The operation delivered significant results: following the cement squeeze job, water production dropped from 900 to 250 barrels per day, while oil output increased substantially, from less than 100 to 250 barrels per day. The water cut was significantly reduced from approximately 90% to 50% (Fig. 5). This marked improvement not only demonstrates the effectiveness of targeted remediation but also highlights the value of integrated diagnostics and corrective techniques in maximizing oil recovery. The outcome represents a key milestone in optimizing well performance and operational efficiency.

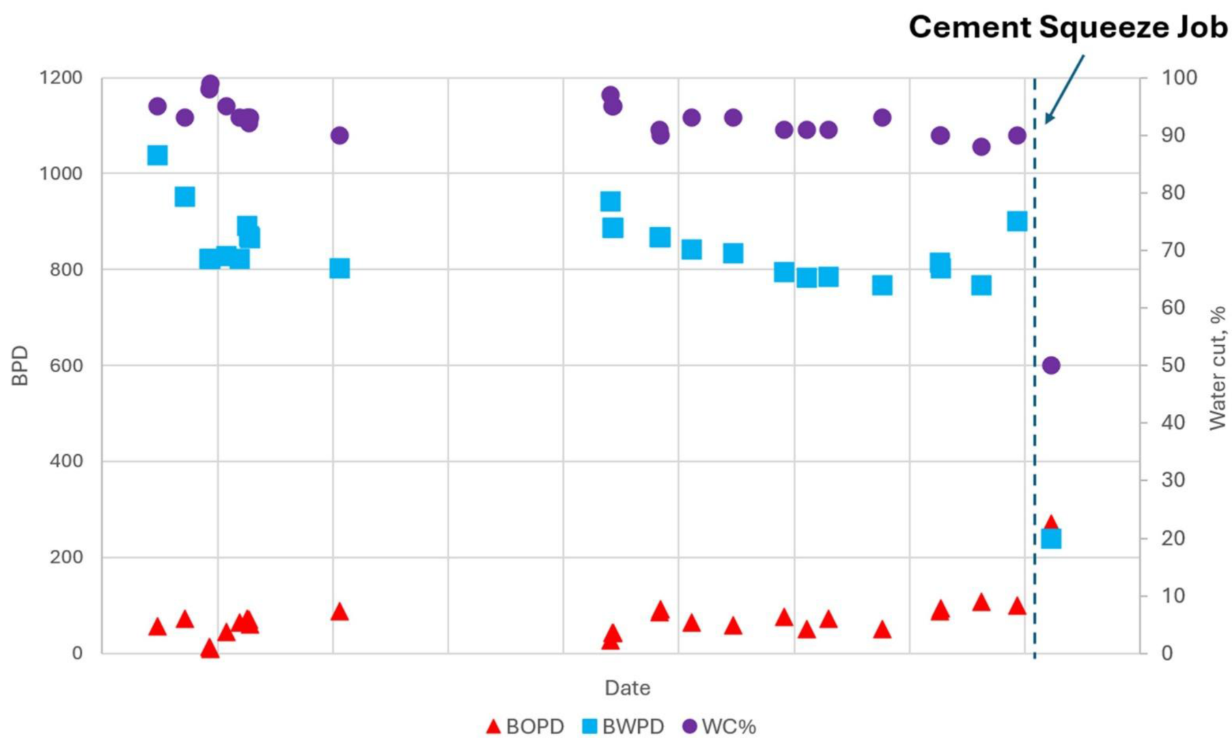

Figure 5—Water Cut Analysis – Cement Squeeze Effectiveness

Conclusions

The integration of passive acoustics, high precision temperature logging, and temperature modelling with the standard PLT approach provides a significantly enhanced understanding of reservoir flow dynamics. In contrast to traditional production logging tools (PLT), which are often limited in detecting flow behind casing or beyond the spinner threshold, this integrated diagnostic approach offers significantly greater resolution and interpretative depth. By using this advanced set of technologies, the operator was able to accurately identify the root cause of excessive water production, which was attributed to behind-casing crossflow.

Passive acoustics detected the distinct acoustic signatures of fluid movement behind the casing. Multichannel spectral noise logging offers high-frequency resolution across a wide spectral range, enabling precise detection and characterization of acoustic signals linked to active inflows and crossflow events within the reservoir [7]. Additionally, the depth-correlated spectral noise drift (SND) technique has proven effective in identifying lateral flows and separating them from wellbore-related noise interference [8].

From another perspective, high-precision temperature logging revealed active flow zones below the perforation interval. The temperature flow simulator (Cascade), based on heat transfer equations and applicable to both injection and production wells, performs detailed numerical simulations of wellbore temperature responses to fluid inflow and behind-casing flow scenarios [9]. This temperature modelling capability was instrumental in quantifying flow volumes originating below the perforation interval, data typically inaccessible through conventional diagnostic methods.

This comprehensive interpretation provided an in-depth view of reservoir conditions, uncovering hidden zones of liquid entry and enabling the development of a targeted and effective remediation strategy. Based on these insights, a cement squeeze operation was carried out to seal the crossflow pathway behind the casing. This intervention directly targeted the primary source of water production, resulting in a marked improvement in well performance: water output decreased from 900 to 250 barrels per day, oil production increased from less than 100 to 250 barrels per day, and the water cut dropped from about 90% to 50%.

These outcomes underscore the value of combining advanced diagnostics with targeted remedial actions. This case clearly demonstrates how integrated reservoir surveillance can resolve complex flow challenges, maximize hydrocarbon recovery, enhance well productivity, and support more efficient reservoir management. Ultimately, this success represents a key milestone in the ongoing optimization of well performance and operational efficiency.

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Nomenclature

APS: Acoustic Power Spectrum
Cascade: Temperature Flow Simulator
CBL: Cement Bond Log
Chorus: Passive Acoustic Logging
ESP: Electric Submersible Pump
GR: Gamma Ray
HPT: High Precision Temperature
LF: Low Frequency
OH: Open Hole Log
PLT: Production Logging Tool
PVT: Pressure Volume Temperature Data
Pres: Pressure
SND: Spectral Noise Drift
TDS: Total Dissolved Solids

Temp: Temperature
Y-tool: A diverter fitting installed above the ESP motor that splits the flow path into two branches: one for the ESP and one for tool access (typically for wireline or slickline)

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